

Raspberry Pi Onramp

A Beginner's Guide to Tinkering with Hardware and Code

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1. General Overview

1. General Overview

Why Use Raspberry Pi

Raspberry Pi is a popular choice for beginners because it is affordable, well-documented, and has a large community. It ships with a familiar Debian-based environment, making it easy to get started with Linux. It is equally suited for advanced users who want to build custom hardware projects, run home servers, or experiment with electronics via its 40-pin GPIO header.

1. General Overview

What is Raspberry Pi

Raspberry Pi is a **low-cost** operating systems. It is a popular platform for single-board computer *capable of running* Linux-based embedded systems, IoT projects, home automation, education, and prototyping. It was originally developed by the Raspberry Pi Foundation (now Raspberry Pi Ltd.) with the goal of promoting computer science education.

The official operating system is **Raspberry Pi OS** (formerly called Raspbian until May 2020). It is based on the Debian Linux distribution and uses the apt (Advanced Package Tool) package manager.

1. General Overview

Before installing any software, it is good practice to refresh the local package index and then apply all available upgrades:

```
1 sudo apt update # refresh the package list
2 sudo apt full-upgrade # upgrade packages, handling dependency changes
```



full-upgrade is preferred over upgrade because it correctly handles cases where packages need to be added or removed to satisfy updated dependencies.

To install a specific package:

```
1 sudo apt install git # install git
```



SSH

Secure Shell Configuration

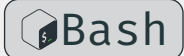
1.1 SSH Setup

SSH (Secure Shell) is a cryptographic network protocol for secure remote access to a machine over an unsecured network. On Raspberry Pi OS, the `openssh-server` package is already installed but **disabled by default** for security reasons.

There are three common ways to enable it:

Option 1 — `raspi-config` (recommended for interactive use):

```
1 sudo raspi-config
2 # Navigate to: Interface Options → SSH → Enable
```



1.1 SSH Setup

Option 2 — systemctl (enable on a running system):

```
1 sudo systemctl enable ssh # enable SSH to start on boot
2 sudo systemctl start ssh # start the SSH service immediately
```



Option 3 — headless setup (before first boot):

Create an empty file named `ssh` (no extension) in the `/boot` or `/boot fs` partition of the SD card. The OS will detect it on first boot, enable the SSH service, and delete the file.

Once SSH is enabled, connect from another machine with:

```
1 ssh <username>@<hostname>.local
2 # e.g. ssh pi@raspberrypi.local
```

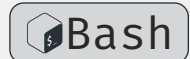


1.1 SSH Setup

INFO

Create a new connection profile of type Ethernet on the physical interface `enp4s0`. The PC will act as a DHCP server and automatically assign an IP address to any device connected via Ethernet (the Pi), while also enabling NAT to share the laptop's internet connection.

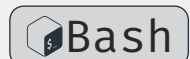
```
1 nmcli connection add type ethernet ifname enp4s0 con-name  
  "rpi-share" ipv4.method shared
```



i

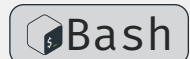
Activate and inspect the connection

```
1 nmcli connection up "rpi-share"  
2 nmcli connection show "rpi-share"
```



Perform a ping sweep across all 254 addresses, without port scanning

```
1 nmap -sn 10.42.0.0/24
```



1.1 SSH Setup

1.1.1 SSH Key Authentication

Connecting with a password is convenient but less secure. The recommended approach is **public-key authentication**: you generate a key pair on your client machine, then place the public key on the Raspberry Pi. From that point on, no password is needed and brute-force attacks become ineffective.

1.1.2 How it works

A key pair consists of two files:

Private key (`~/.ssh/rpi`) stays on your machine, never shared.

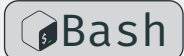
Public key (`~/.ssh/rpi.pub`) copied to the Pi. It can only be used to verify someone who holds the matching private key.

1.1 SSH Setup

1.1.3 Generate a key pair (on your local machine)

Ed25519 is the current recommended algorithm: it is compact, fast, and more secure than RSA.

```
1 ssh-keygen -t ed25519 -C "your_email@example.com"
```



You will be prompted for:

File location press Enter to accept the default (`~/.ssh/rpi`).

Passphrase strongly recommended. It encrypts the private key on disk so that even if the file is stolen, it cannot be used without the passphrase.

The command produces two files:

```
1 ~/.ssh/rpi      # private key – keep this secret
2 ~/.ssh/rpi.pub  # public key  – this goes on the Pi
```

1.1.4 Copy the public key to the Raspberry Pi

The simplest method is `ssh-copy-id`, which handles directory creation and permissions automatically:

1.1 SSH Setup

```
1  ssh-copy-id <username>@<pi-ip-address>  
2  # e.g. ssh-copy-id pi@10.42.0.39
```



You will be asked for your Pi password one final time. After this step, the public key is appended to `~/.ssh/authorized_keys` on the Pi.

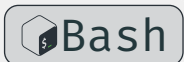
1.1.5 Connect using the key

```
1  ssh <username>@<pi-ip-address>  
2  # e.g. ssh pi@192.168.1.123
```



SSH automatically finds and uses `~/.ssh/id_ed25519`. If you set a passphrase, you will be prompted for it once. To avoid re-entering it every session, add the key to the SSH agent:

```
1  eval "$(ssh-agent -s)" # start the agent  
2  ssh-add ~/.ssh/rpi # load the key (prompts for passphrase once)
```



1.1 SSH Setup

1.1.6 Disable password login on the Pi

Once key authentication is confirmed to be working, you can disable password logins entirely to harden the Pi against brute-force attacks. Edit the SSH server config:

```
1 sudo vim /etc/ssh/sshd_config
```



Find and set the following lines (*restart the ssh service afterwards*):

```
1 PasswordAuthentication no
2 PubkeyAuthentication yes
```

Then restart the SSH service:

```
1 sudo systemctl restart ssh
```



WARNING



Do not disable password login until you have verified that key authentication works in a separate terminal session. Locking yourself out would require physical access to the Pi to recover.

Git and GitHub

Version Control and Collaboration

1.2 Git and GitHub

Git is a distributed version control system that tracks changes to files and directories over time. It allows you to commit snapshots of your work, branch off for experiments, and merge changes back together.

GitHub is a cloud-based hosting platform for Git repositories. It adds collaboration features such as pull requests, issues, and Actions on top of standard Git.

To install Git:

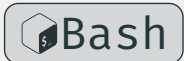
```
1 sudo apt install git
```



1.2 Git and GitHub

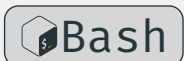
Basic initial configuration:

```
1 git config --global user.name "Your Name"
2 git config --global user.email "you@example.com"
```



To clone an existing repository from GitHub:

```
1 git clone https://github.com/a-mhamdi/rpi-onramp.git
```



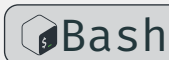
1.2 Git and GitHub

Using a key with GitHub

The same key pair `rpi` and `rpi.pub` can authenticate you to GitHub, removing the need for HTTPS tokens when pushing code.

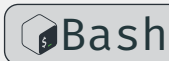
Display your public key and copy the output. Then go to GitHub → Settings → SSH and GPG keys → New SSH key, paste it in, and save. Test the connection:

```
1 ssh -T git@github.com
2 # Expected: Hi <username>! You've successfully authenticated ...
```



To switch an existing local repository from HTTPS to SSH:

```
1 git remote set-url origin git@github.com:<user>/<repo>.git
```



1.2 Git and GitHub

Lazygit

Lazygit is a terminal UI for Git that makes complex operations simple — stage individual lines, perform interactive rebases, and cherry-pick commits, all without memorising obscure flags. It runs directly in your terminal, so there is no need to switch between your editor and a separate GUI application. It is especially useful on the Raspberry Pi where you are often working entirely over SSH with no desktop environment.

To install it on Raspberry Pi OS:

```
1 sudo apt install lazygit
```



Launch it from inside any Git repository by simply running `lazygit`. Press `?` at any time to view the full list of keybindings.

1.2 Git and GitHub

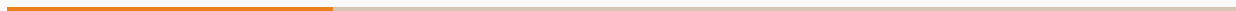
The screenshot shows the LazyGit interface with the following sections:

- [1]-Status**: Shows the current branch as `main` and the working directory as `RPI-Onramp`.
- [2]-Files**: A tree view of the project files. The `parts` directory is expanded, showing `ec.typ` (selected), `py.typ`, `rpi.typ`, `common.typ`, `main.pdf`, `main.typ`, and `blink.py`.
- [3]-Local branches - Remotes - Tags**: Shows the `main` branch as the current branch.
- [4]-Commits - Reflog**: Shows a list of commits, with the most recent one being `053a0a5a` (update README file).
- [5]-Stash**: Shows no stashed changes.
- Unstaged changes**: A diff view showing the changes in `b/docs/parts/ec.typ`. The diff indicates a new file mode and the addition of content for basic circuitry, UART, analog inputs-outputs, digital inputs-outputs, I2C, and SPI.
- Command log**: Shows the command `be asked if you want to force push`.

At the bottom, there is a status bar with the following text: `Stage: <space> | Commit: c | Edit: e | Stash: s | Discard: d | Reset: D | Keybindings: ? | ... Donate Ask Question 0.47.2`

For full documentation and source code, see github.com/jesseduffield/lazygit.

2. Julia Codes



2. Julia Codes

Julia is a high-level, high-performance programming language designed for numerical and scientific computing. While it is often associated with workstations and servers, Julia runs well on ARM-based hardware including the **Raspberry Pi**. This makes it an attractive choice for edge computing, data logging, sensor fusion, and lightweight network services running directly on embedded Linux systems.

This document covers how to install Julia on a **Raspberry Pi**, how to manage GPIO and serial connections, how to use Julia's networking stack, and how to write practical programs that tie these capabilities together.

Getting Started with Julia on Raspberry Pi

Installation and First Steps

2.1 Initial Setup

2.1.1 Hardware Requirements

Julia can run on any Raspberry Pi model that supports a 64-bit ARM operating system, starting from the Raspberry Pi 3. The Raspberry Pi 4 or 5 is recommended for interactive use and package compilation due to its faster CPU and larger RAM. A microSD card of at least 16 GB is advisable, as Julia's package depot and precompiled artifacts can occupy several gigabytes.

2.1 Initial Setup

2.1.2 Operating System

Install the 64-bit version of Raspberry Pi OS (formerly Raspbian). Julia's official ARM builds target aarch64, so the 64-bit OS is required to use them directly. You can download the image from the Raspberry Pi website and flash it with Raspberry Pi Imager.

After first boot, run a full system update before installing Julia:

```
1 sudo apt update && sudo apt upgrade -y
```



2.1 Initial Setup

2.1.3 Installing Julia

The recommended way to install Julia on Raspberry Pi is through `juliaup`, the official Julia version manager.

```
1 curl -fsSL https://install.julialang.org | sh
```



Follow the prompts. After installation, open a new shell session or source your profile:

```
1 source ~/.bashrc
```



Verify the installation:

```
1 julia --version
```



2.1 Initial Setup

juliaup allows you to maintain multiple Julia versions and switch between them with

```
1 juliaup default <version>
```



INFO

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If you prefer a simpler path without juliaup, the Raspberry Pi OS repositories include a Julia package, though it may be an older version:

```
1 sudo apt install julia
```



Using the Julia REPL

Interactive Programming and Package Management

2.2 First Launch and the REPL

Start Julia by typing `julia` in the terminal. You will be greeted by the interactive REPL (*Read-Eval-Print Loop*). From here you can enter expressions, manage packages, and run scripts.

Julia has four REPL modes:

Julia mode (default) execute Julia expressions.

Package mode (press `]`) manage packages with `add`, `rm`, `status`, `update`.

Help mode (press `?`) display documentation for any symbol.

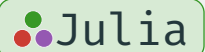
Shell mode (press `;`) run shell commands without leaving Julia.

2.2 First Launch and the REPL

2.2.1 Package Depot and Precompilation

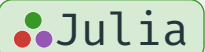
To install a package, enter package mode and type:

```
1 ] add Plots
```



To exit the REPL:

```
1 exit()
```



Julia compiles packages on first use. On a Raspberry Pi this can be slow — several minutes for large packages — but subsequent launches are fast. Precompilation happens automatically when you add a package or using it for the first time.

2.2.2 Julia Onramp

 Source code is available at [codes/julia-onramp.ipynb](https://github.com/JuliaLang/julia-onramp/blob/master/README.md)

2.2 First Launch and the REPL

2.2.3 Application: Basic Backprop

- An implementation of backpropagation for a simple perceptron.
- Gradients computation and weights update using Julia's basic operations.



Source code is available at [codes/basic-backprop.jl](https://github.com/JuliaLang/julia/blob/master/src/basic-backprop.jl)

```
(_)  Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.12.6 (2026-04-09)
Official https://julialang.org release

(@v1.12) pkg> activate .
Activating project at `~/rpi-onramp/codes`

julia> include("basic-backprop.jl")
-1.25
-1.998206898724217 0.1492029387806452

1.35839 data
-0.0395647
-298.97 10300.0

julia> |
```

3. Basic Circuitry

Digital I/O

Interfacing with the Physical World

3.1 Digital Inputs-outputs

The Raspberry Pi exposes **40 GPIO pins** (*on models B+ and later*), each configurable as either **input** or **output** at runtime.

Output mode

- Drive pin HIGH (3.3 V) or LOW (0 V)
- Power LEDs, relays, transistors
- Max current: **16 mA** per pin

Input mode

- Read external logic levels
- Optional pull-up / pull-down resistors
- Voltage-safe range: 0 – 3.3 V

WARNING



GPIO operates at 3.3 V logic — never connect 5 V signals directly.

3.1 Digital Inputs-outputs

Julia accesses GPIO through the **PiGPIO.jl** package, which wraps the pigpio C daemon running on the Pi.

 Source code is available at [codes/digital-io.jl](https://github.com/JuliaGPIO/JuliaGPIO.jl)

NOTE

Requires pigpiod running: `sudo pigpiod` · Install: `]add PiGPIO`

Analog I/O

Measuring and Controlling Continuous Signals

3.2 Analog Inputs-outputs

Unlike Arduino, the Raspberry Pi has **no built-in ADC** — all GPIO pins are strictly digital. Analog I/O requires external hardware.

Analog Input (ADC)

- Common chip: **MCP3008** (10-bit, 8-ch)
- Communicates over **SPI**
- Resolution: 1024 steps over 0 – 3.3 V
- 3.2 mV per step

Analog Output (DAC)

- Common chip: **MCP4725** (12-bit)
- Communicates over **I²C**
- Resolution: 4096 steps over 0 – 3.3 V
- 0.8 mV per step

3.2 Analog Inputs-outputs

PiGPIO.jl provides hardware-timed PWM on any GPIO pin via the pigpiod daemon, giving clean pulses without CPU spin.

INFO

i

PWM can approximate analog output without a DAC — useful for motor speed and LED dimming.

 Source code is available at [codes/analog-io.jl](https://github.com/JuliaGPIO/JuliaGPIO.jl/blob/master/src/pwm.jl)

Line Encoding Schemes

NRZ-L · NRZ-I · Manchester · Differential Manchester

3.3 Line Encoding

Line encoding is how binary data (0s and 1s) gets mapped to a **physical signal** on a wire — defined by voltage levels and when they change.

Voltage Level

High (+V) or Low (-V / 0V) represents a bit

Transitions

A signal change (edge) can itself carry meaning

Clock Recovery

Receiver needs to stay in sync with the sender

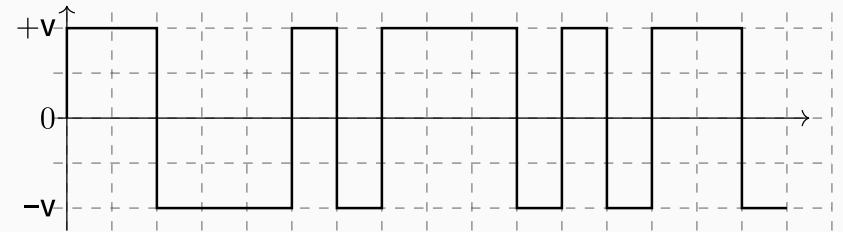
Different schemes trade off between **simplicity**, **bandwidth**, and **self-clocking**.

3.3 Line Encoding

NRZ-L — Non-Return to Zero Level

Rule: Level directly represents the bit value.

- **High** (+V) → bit **0**
- **Low** (-V) → bit **1**
- Signal stays at that level for the entire bit period
- No transition needed — hence “non-return to zero”



Advantage: Simple, efficient use of bandwidth. (Used in RS-232)

Problem: Long runs of the same bit produce no transitions → receiver loses clock sync.

3.3 Line Encoding

NRZ-I — Non-Return to Zero Inverted

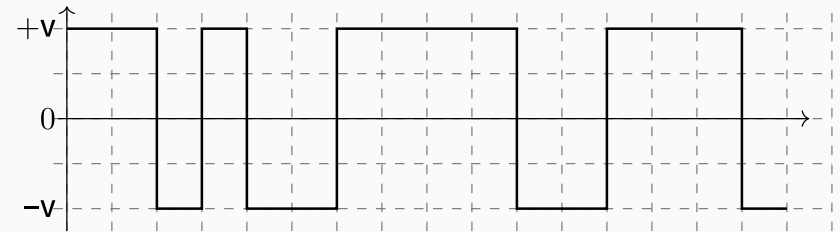
Rule: A **transition** encodes the bit, not the level.

- Bit **1** → **transition** at the start of the bit period
- Bit **0** → **no transition** (level stays the same)

This is **differential** coding: the meaning depends on change, not absolute voltage.

Advantage: Immune to polarity inversion (swapped wires don't corrupt data). (Used in USB, HDLC)

Problem: Long runs of **0s** still produce no transitions → still loses sync.



3.3 Line Encoding

Manchester Encoding

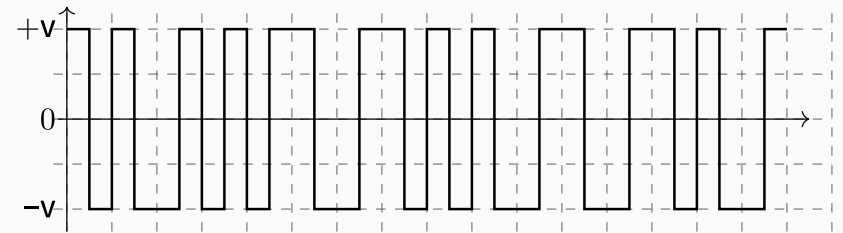
Rule: Each bit period is split in half — the bit is encoded by the **mid-bit transition direction**.

- Bit **0** → **High→Low** transition at mid-bit
- Bit **1** → **Low→High** transition at mid-bit

Every bit has a transition in the middle → **self-clocking**: receiver can always recover the clock.

Advantage: No long runs without transitions, no DC component. (Used in 10BASE-T Ethernet, IR remotes)

Cost: Requires **2× the bandwidth** of NRZ — each bit takes two signal intervals.



3.3 Line Encoding

Differential Manchester Encoding

Rule: Mid-bit transition is **always** present (*for clocking*).

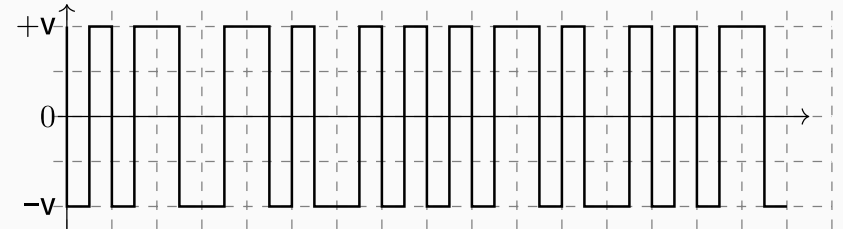
The bit value is encoded by what happens **at the start**:

- Bit **0** → **transition** at the start of the bit period
- Bit **1** → **no transition** at the start

Combines ideas from NRZI (*differential*) and Manchester (*guaranteed mid-bit edge*).

Advantages: Always self-clocking (mid-bit edge). Immune to polarity inversion. (Used in Token Ring, IEEE 802.5)

Cost: Same 2× bandwidth overhead as Manchester.



3.3 Line Encoding

4B/5B — Block Encoding

Rule: Every group of **4 data bits** is replaced by a **5-bit code word** chosen to guarantee enough transitions.

The 5-bit codes are picked so no code has more than **one leading zero** or **two trailing zeros** — ensuring NRZI transitions happen frequently.

INFO

i 4B/5B is always paired with NRZI — it's a pre-coding step, not a standalone scheme.

1	4-bit	5-bit
2	————	————
3	0000	11110
4	0001	01001
5	0010	10100
6	...	
7	1111	11101
8		
9	Unused codes → control symbols (idle, start, end)	

3.3 Line Encoding

Summary Comparison

Scheme	Self-Clocking	DC-Free	BW Overhead	Used In
NRZ-L	✗	✗	0%	RS-232
NRZ-I	Partial	✗	0%	USB, HDLC
Manchester	✓	✓	100%	10BASE-T, IR
Diff. Manchester	✓	✓	100%	Token Ring
4B/5B + NRZI	✓	Partial	25%	Fast Ethernet

UART and RS-232

Asynchronous Serial Communication Protocols

3.4 UART & RS-232

What is UART?

UART — Universal Asynchronous Receiver/Transmitter — is a hardware protocol for serial communication between two devices.

Asynchronous

No shared clock line. Both sides agree on speed in advance.

Serial

Bits sent one at a time over a single wire per direction.

Full-Duplex

TX and RX are separate wires — both sides can talk at once.

UART is a **logic-level protocol**. RS-232 is one electrical standard built on top of it — defining voltages, connectors, and cable lengths.

3.4 UART & RS-232

UART Frame Structure

A UART frame wraps each byte with control bits:

Idle line sits **High** when nothing is sent

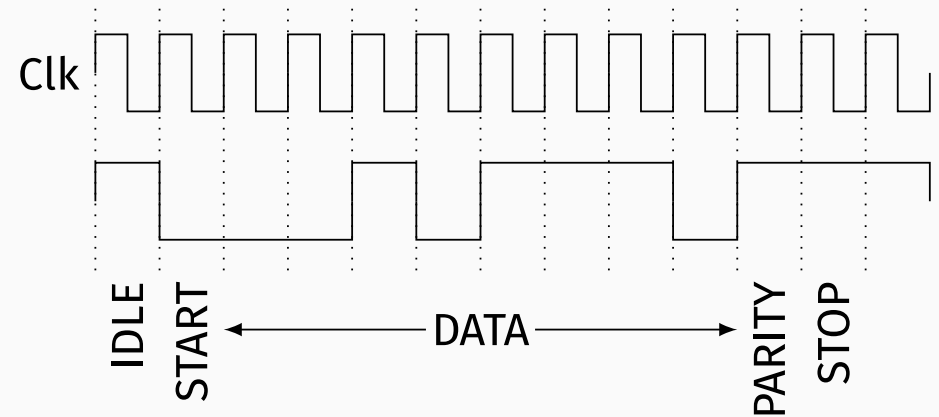
Start bit always **0** (Low), signals frame start

Data bits 5 to 9 bits, LSB first (typically 8)

Parity bit optional error check (even/odd/none)

Stop bit(s) always **1** (High), 1 or 2 bits

The receiver detects the **High→Low** edge of the start bit, then samples each bit at the center of its period.



3.4 UART & RS-232

Baud Rate

Baud rate = number of **symbols per second** on the line.
For UART (one bit per symbol): baud rate = bit rate.

Both sides must be configured to the **same baud rate** — there is no negotiation. A mismatch causes garbage data.

Common standard rates:

Baud Rate	Bit Period
9 600	104.2 μ s
19 200	52.1 μ s
115 200	8.68 μ s
921 600	1.08 μ s

UART **tolerates** a clock mismatch of up to **$\pm 2-3\%$** before bit sampling drifts enough to cause errors. Over a 10-bit frame, even a 2% error shifts the sample point by **0.2 bit periods** by the last bit.

3.4 UART & RS-232

Parity — Simple Error Detection

The optional **parity bit** is set so the total number of 1-bits in the frame (data + parity) satisfies a rule:

Even parity total count of 1s is even

Odd parity total count of 1s is odd

None no parity bit, faster transmission

Detects: any **single-bit** error.

Misses: any **even number** of flipped bits.

```
1 Data: 0b10110010
2 1-bits: 4 (even)
3
4 Even parity bit → 0
5 (total 1s stays 4)
6
7 Odd parity bit → 1
8 (total 1s = 5)
```

Parity does **not** correct errors — only flags them. For correction, higher-level protocols are needed.

3.4 UART & RS-232

RS-232 — Voltage Levels

RS-232 uses **inverted** and **wide-swing** voltages compared to logic levels:

Logic 1 (Mark / Idle)

Voltage between **-3 V and -15 V**
Typically **-12 V** in practice

Logic 0 (Space / Start)

Voltage between **+3 V and +15 V**
Typically **+12 V** in practice

The **±3 V dead zone** (-3 V to +3 V) is undefined — signals in this range are ignored, giving noise immunity.

Key point: Logic 1 = negative voltage. This is **opposite** to TTL/CMOS logic (where 1 = High). A **level shifter** (e.g. MAX232) is required to interface RS-232 with a microcontroller.

3.4 UART & RS-232

RS-232 — DB9 Connector & Signals

The classic RS-232 connector is the **DE-9** (often called DB9).

Pin	Name	Function
1	DCD	Data Carrier Detect
2	RXD	Receive Data
3	TXD	Transmit Data
4	DTR	Data Terminal Ready
5	GND	Signal Ground
6	DSR	Data Set Ready
7	RTS	Request To Send
8	CTS	Clear To Send
9	RI	Ring Indicator

For a **minimal connection** (*no flow control*), only **3 wires** are needed:

Device A

Device B

TXD ————— RXD

RXD ————— TXD

GND ————— GND

RTS/CTS add hardware flow control — the sender checks CTS before transmitting to avoid overflowing the receiver.

3.4 UART & RS-232

UART vs RS-232 — What's the Difference?

Aspect	UART	RS-232
What it is	Logic protocol	Electrical standard
Voltage (logic 1)	+3.3 V or +5 V	-3 V to -15 V
Voltage (logic 0)	0 V	+3 V to +15 V
Max cable length	Short (PCB traces)	15 m at 9600 baud
Noise immunity	Low	High (wide swing)
Connector	None defined	DE-9 / DE-25
Interface chip	None needed	MAX232 or similar

- ◆ UART logic levels are for chip-to-chip communication on a PCB.
- ◆ RS-232 was designed for long cables in industrial environments.

3.4 UART & RS-232

 Source code is available at [codes/tx-rx.jl](https://github.com/mhamdi/codes/blob/master/tx-rx.jl)

Inter-integrated Circuits

A Multi-Master, Multi-Slave, Packet-Switched, Single-Ended, Serial Communication Bus

What is I²C?

I²C is a **synchronous, half-duplex, multi-master** serial bus developed by Philips in 1982. Only **2 wires** are needed regardless of how many devices are on the bus.

2 Wires Only

SDA (data) and SCL (clock)
— shared by all devices
on the bus.

Addressed

Every slave has a 7-bit
(or 10-bit) address. No CS
pins needed.

ACK/NACK

Slave acknowledges
every byte — master
knows if data arrived.

Multi-Master

Multiple masters can
share the bus with built-
in arbitration.

Common uses: sensors (temp, IMU, humidity), EEPROMs, RTCs, LCD controllers, power management ICs.

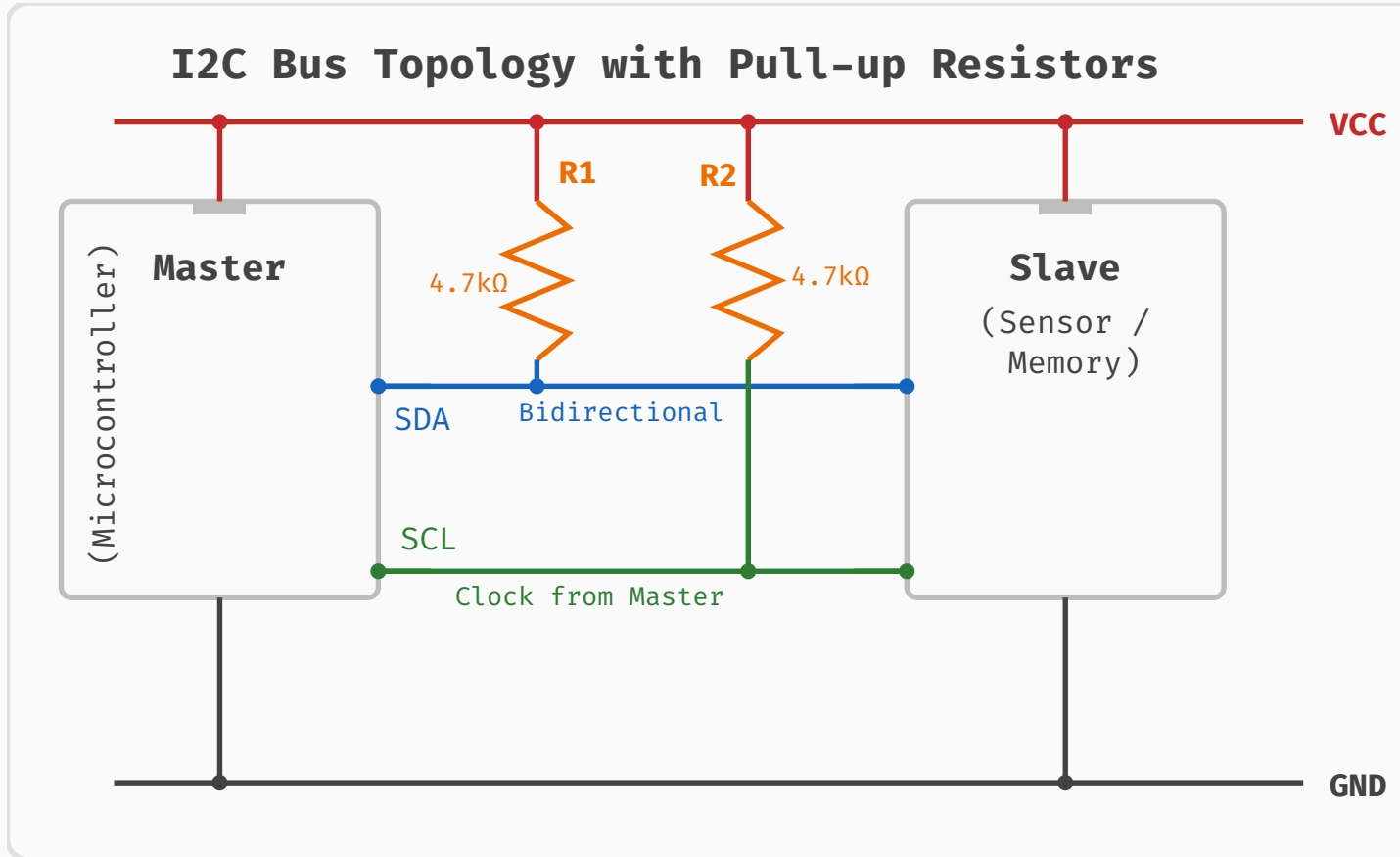
Bus Topology & Pull-up Resistors

All devices share the same SDA and SCL wires. SDA and SCL are **open-drain** lines — devices can only pull them **Low**. Pull-up resistors connected to VCC bring the lines **High** when nobody is pulling them down.

This allows any device to signal without short-circuiting others.

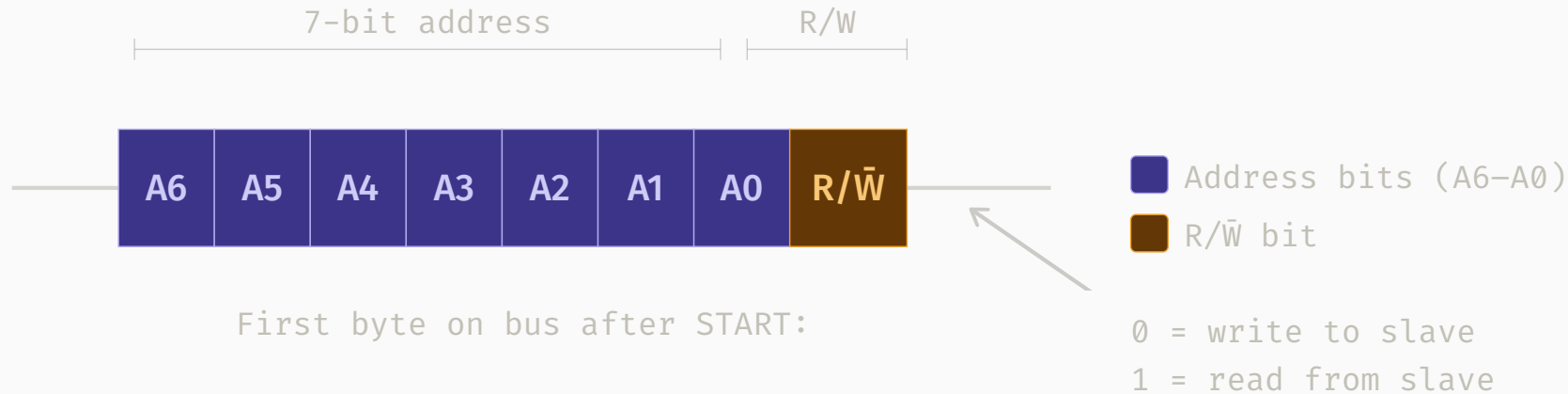
- ♦ **Too high** → slow rise time, signal corruption at high speed.
- ♦ **Too low** → excessive current, devices struggle to pull Low.

3.5 I2C



Device Addressing

Every slave has a **7-bit address** (most common) hardwired or configurable via address pins.

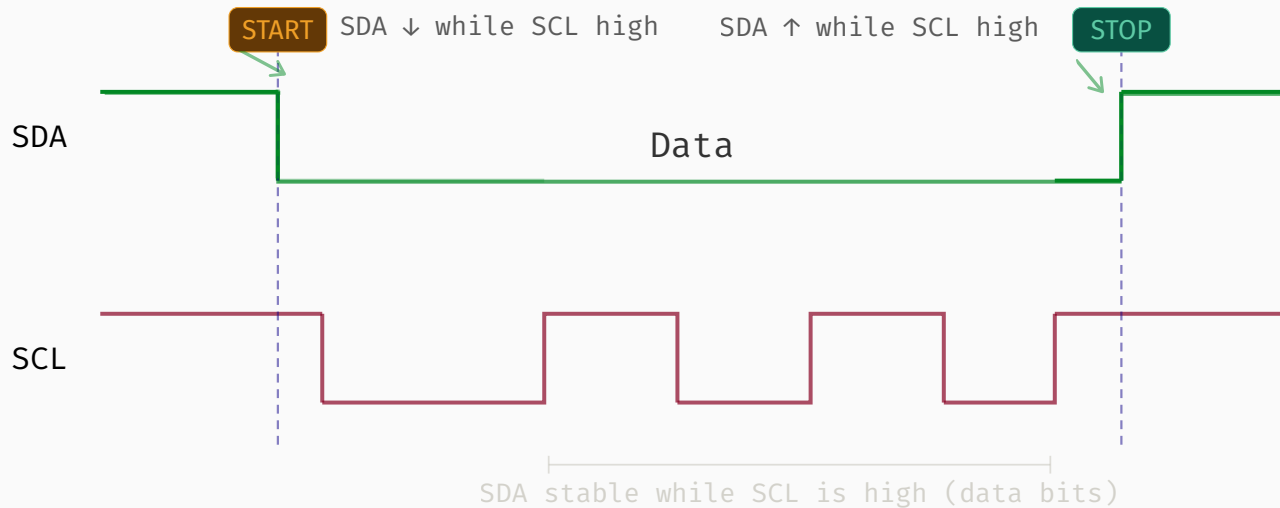


The master sends the address at the start of every transaction, followed by a **R/ $\bar{W}$ bit**:

- **0** → master wants to **write** to slave
- **1** → master wants to **read** from slave

START and STOP Conditions

I²C uses special **bus conditions** — not regular data bits — to delimit transactions. They are unique because SDA changes **while SCL is High** (normally forbidden during data transfer).



START (S)

SDA goes **High→Low** while SCL is **High**.
Signals the beginning of a transaction.
Bus is now "busy".

STOP (P)

SDA goes **Low→High** while SCL is **High**.
Signals end of transaction.
Bus returns to "idle".

Data Transfer & ACK/NACK

Data is transferred **8 bits at a time**, MSB first. After each byte, there is a mandatory **9th clock pulse** for the acknowledgement bit:

ACK (0) receiver pulls SDA **Low**: “got it, send more”

NACK (1) SDA stays **High**: “stop” or “not ready”

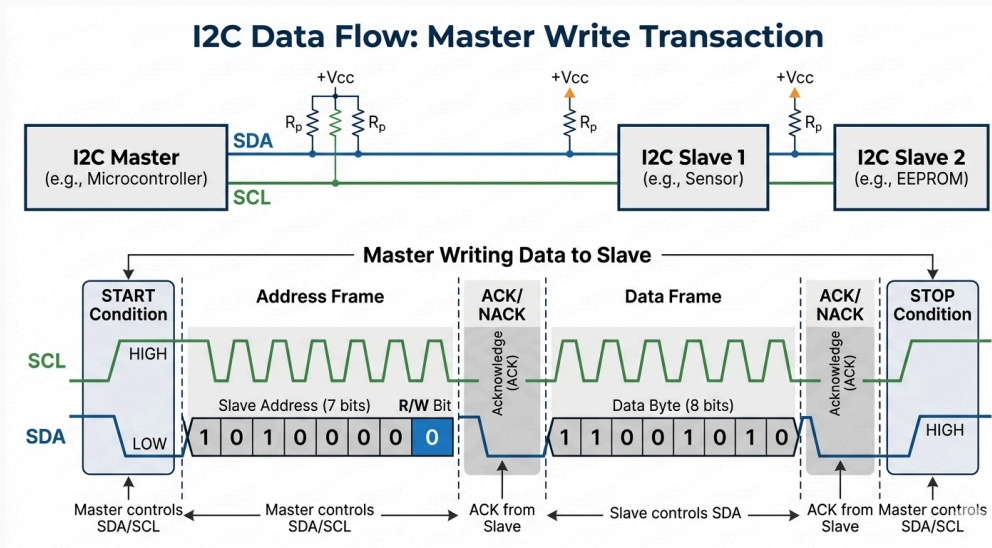
Who sends ACK?

Write transaction slave ACKs each byte from master

Read transaction master ACKs each byte from slave; master sends NACK on the **last** byte to signal it's done

Write Transaction

A typical register read (e.g. from a sensor) uses a **Repeated START**:



Phase 1 — Write

Master sends slave address + register it wants to read from.

Repeated START

Bus stays busy. Master switches direction without a STOP.

Phase 2 — Read

Master re-addresses slave in Read mode. Slave drives SDA.

Clock Stretching

A slave that needs more time to process data can **hold SCL Low** after the master releases it — effectively pausing the clock.

The master, before driving SCL High again, checks if the line is already High. If it finds SCL still Low (slave holding it), it waits until the slave releases it.

Use cases:

- Slow ADC completing a conversion
- EEPROM finishing an internal write cycle
- Slave needing time to prepare the next data byte

Risk: A buggy slave that never releases SCL will **hang the bus**. Some masters implement a timeout and reset.

IMPORTANT



Only possible because SCL is open-drain — slave can pull Low without fighting the master.

I²C Speed Modes

Mode	Max Speed	Pull-up	Notes
Standard Mode (Sm)	100 kHz	4.7 kΩ	Original spec, most devices support
Fast Mode (Fm)	400 kHz	2.2 kΩ	Very widely supported
Fast-Mode Plus (Fm+)	1 MHz	1 kΩ	Requires stronger drivers
High-Speed Mode (Hs)	3.4 MHz	special	Special master code, rare
Ultra-Fast Mode (UFm)	5 MHz	N/A	Unidirectional, no ACK, very rare

INFO

i

Most embedded projects use **Standard** or **Fast** mode. Not all slaves support Fast-Mode Plus or above — always check the datasheet. Higher speeds require shorter bus lengths and lower pull-up values.

I²C — Pros & Cons

- **Only 2 wires** regardless of slave count
- Built-in **ACK/NACK** — detects missing slaves
- **Multi-master** with hardware arbitration
- **Clock stretching** for slow slaves
- Simple board routing, long bus possible at low speed

- **Half-duplex** — can't read and write simultaneously
- **Slower** than SPI (max 5 MHz vs 100 MHz)
- **Address conflicts** possible with fixed-address slaves
- Open-drain + pull-ups add **capacitance** — limits bus length at high speed
- Protocol overhead (address + ACK per byte)

 Source code is available at [codes/i2c.jl](https://github.com/mhamdi/i2c.jl)

Serial Peripheral Interface

A High-Speed Synchronous Communication Protocol

3.6 SPI

What is SPI?

SPI is a **synchronous, full-duplex, master-slave** serial bus. Unlike UART, a shared **clock line** is provided by the master — no baud rate agreement needed.

Synchronous

Clock driven by master.
Both sides latch on the same edge.

Full-Duplex

Data flows in both directions simultaneously on separate wires.

Master-Slave

Master controls the clock and selects which slave talks.

No Addressing

Slaves are selected by a dedicated chip select (CS) line.

Common uses: flash memory, SD cards, ADCs/DACs, displays, sensors.

3.6 SPI

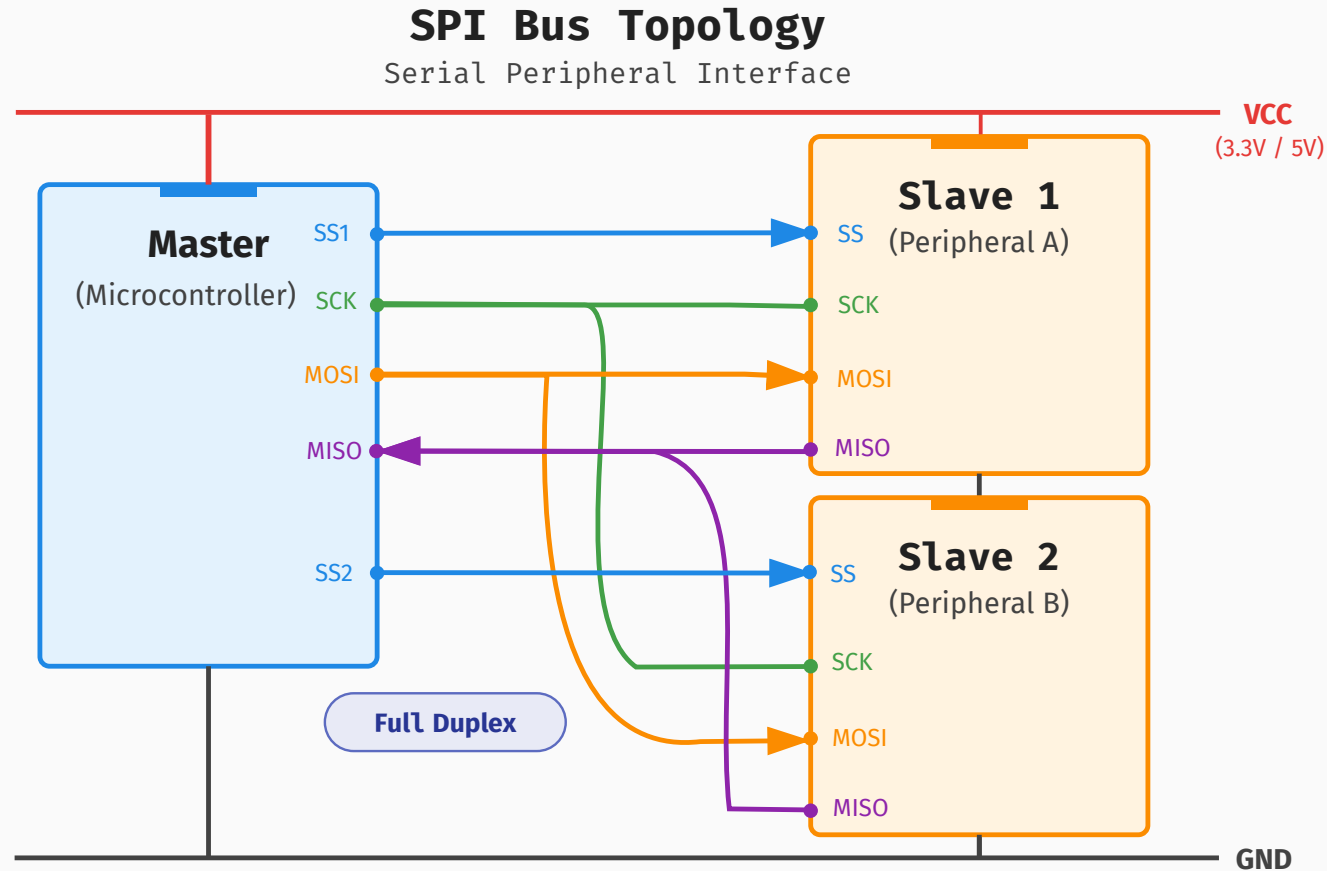
SPI Signal Lines

SPI uses **4 wires** (plus ground):

Signal	Dir	Description
SCLK	M→S	Serial Clock — generated by master
MOSI	M→S	Master Out Slave In — data to slave
MISO	S→M	Master In Slave Out — data from slave
CS/SS	M→S	Chip Select — active low , selects slave

- ◆ Some datasheets use SDI/SDO or DIN/DOUT instead of MOSI/MISO.
- ◆ CS is sometimes called SS (Slave Select) or NSS (active low implied).

3.6 SPI



Note: Standard SPI uses push-pull drivers; pull-up resistors are generally not required (unlike I2C).

3.6 SPI

How a Transfer Works

1. Master pulls **CS Low** to select the slave
2. Master generates **clock pulses** on SCLK
3. On each clock cycle:
 - Master shifts one bit out on **MOSI**
 - Slave shifts one bit out on **MISO**
1. After all bits, master releases **CS High**

Both devices shift simultaneously through an internal shift register it's essentially a **circular exchange** of bytes.

NOTE



Even if you only want to **read**, you must still clock out dummy bytes on MOSI to generate the clock.

3.6 SPI

SPI Clock Modes (CPOL & CPHA)

Two parameters control timing. Each has 2 options → **4 modes** total.

CPOL — Clock Polarity

CPOL=0 → clock idles **Low**

CPOL=1 → clock idles **High**

CPHA — Clock Phase

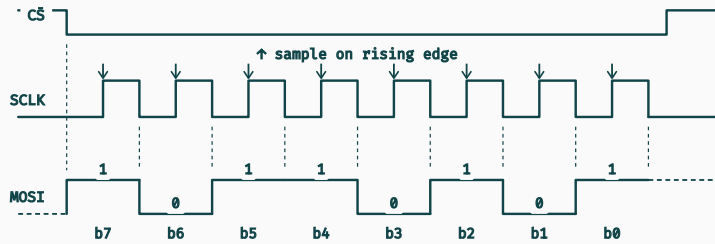
CPHA=0 → data sampled on **leading** edge

CPHA=1 → data sampled on **trailing** edge

Mode	CPOL	CPHA	Idle Clock	Sample On	Common Use
0	0	0	Low	Rising edge	Most sensors, SD cards
1	0	1	Low	Falling edge	Some ADCs
2	1	0	High	Falling edge	Some flash chips
3	1	1	High	Rising edge	Some displays

3.6 SPI

SPI Mode 0 – CPOL=0, CPHA=0

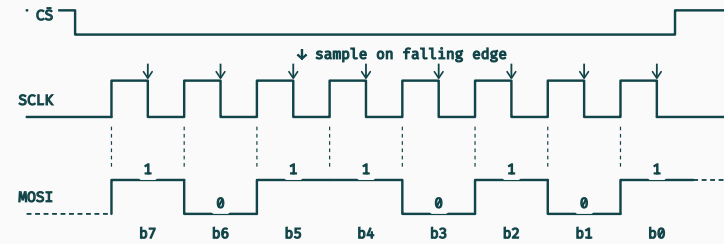


→ Sample (rising edge) ---- Drive next bit (falling edge / CS↓) Hi-Z

CPOL = 0 → clock idles low

CPHA = 0 → first bit driven on CS↓ (before SCLK↑);
subsequent bits on SCLK↓; sampled on SCLK↑

SPI Mode 1 – CPOL=0, CPHA=1

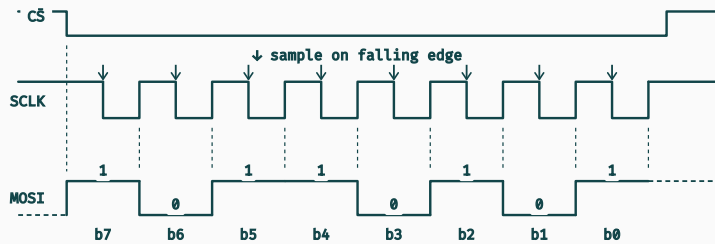


→ Sample (falling edge) ---- Drive next bit (rising edge) Hi-Z

CPOL = 0 → clock idles low

CPHA = 1 → first bit driven on SCLK↑;
data sampled on SCLK↓ (second edge)

SPI Mode 2 – CPOL=1, CPHA=0

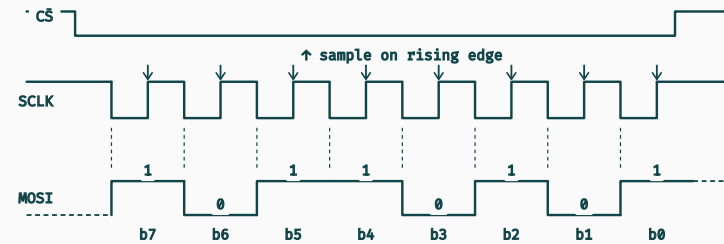


→ Sample (falling edge) ---- Drive next bit (rising edge / CS↓) Hi-Z

CPOL = 1 → clock idles high

CPHA = 0 → first bit driven on CS↓;
data sampled on SCLK↓ (first edge); shifted on SCLK↑

SPI Mode 3 – CPOL=1, CPHA=1



→ Sample (rising edge) ---- Drive next bit (falling edge) Hi-Z

CPOL = 1 → clock idles high

CPHA = 1 → first bit driven on SCLK↓;
data sampled on SCLK↑ (second edge)

Multiple Slaves

Independent CS lines *(most common)*

Each slave has its own CS pin on the master. SCLK, MOSI, MISO are shared. Only one CS is pulled low at a time.

Pro: Simple, fast, independent.

Con: One CS pin per slave on master.

Daisy-chain *(less common)*

Slaves are chained: MISO of one feeds MOSI of the next. A single CS activates all. Data ripples through the chain.

Pro: Only 1 CS line needed.

Con: All slaves must support it; latency increases with chain length.

SPI — Pros & Cons

- **Fast** — few MHz to 100 MHz, no protocol overhead
- **Simple** hardware and firmware
- **Full-duplex** — read and write simultaneously
- No addressing logic — CS line handles selection
- Works with almost any slave device

- **4 wires minimum** — grows with slave count (CS pins)
- No acknowledgement — master can't detect absent slave
- No built-in error detection (no parity, no CRC in protocol)
- Short distances only — no defined max cable length
- No multi-master support in standard spec



Source code is available at [codes/spi.jl](https://github.com/mhamdi/codes/tree/master/spi.jl)

SPI vs UART vs I²C

Feature	SPI	UART	I ² C
Clock	Shared (master)	None (async)	Shared (master)
Wires (2 nodes)	4	2	2
Duplex	Full	Full	Half
Speed	Up to 100 MHz	Up to 1 Mbps	Up to 5 Mbps (FM+)
Multi-slave	CS per slave	Point-to-point	Address per slave
Addressing	Hardware CS	None	7/10-bit address
Hardware cost	Low	Very low	Low
Complexity	Low	Very low	Medium

Thank you for your attention

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